



# Janindu Bandara

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📅 Date of birth: 25.10.1996

## About

- Master's candidate in **Electrical Communication Engineering** at the **University of Kassel**, with a background spanning *embedded systems*, *digital hardware design*, and *signal processing*.
- Hands-on experience programming **microcontrollers** (ATmega328P, ESP32, NodeMCU ESP8266) and designing **digital ICs** in *Verilog HDL*, verified through FPGA implementation and simulation testbenches.
- Conducted master's thesis research in collaboration with **Rohde & Schwarz GmbH & Co. KG**, developing and validating a large-scale MATLAB simulation framework for constrained 5G waveform optimization, reinforcing rigorous design, verification, and documentation practices in a professional engineering environment.
- Practical exposure to *robotics and sensor integration* (Webots, GPS-based navigation), *IoT hardware*, and *ECAD tools* (OrCAD); proficient in *C/C++*, *MATLAB*, and *Python*.

## Experience

### Master Thesis Student

Rohde & Schwarz GmbH & Co. KG

München, Germany

Oct 2025 – April 2026

- Designed **FDSS** pulse filters with **symmetric spectral extension (SE)** for DFT-s-OFDM waveforms, recovering a ~2 dB **BER** penalty caused by per-subcarrier amplitude variation under **MMSE** equalization.
- Performed constrained waveform optimization over a **630-configuration parameter grid** (truncation factor, roll-off, extension ratio), balancing **PAPR**, **ACLR**, and throughput against **3GPP Power Class 3** limits; identified two **BER-optimal** operating regimes across a 0–10 dB SNR range.
- Benchmarked optimized **FDSS** waveform against **CP-OFDM**, **DFT-s-OFDM**, and **OTFS** across **AWGN**, **TDL-D** (indoor, 3 km/h), and **TDL-C** (highway, 120 km/h) 3GPP channel models; achieved **PAPR** as low as 2.87 dB while satisfying **ACLR**  $\geq 30$  dBc.
- **Tools & Systems Exposure:** *MATLAB (5G NR Toolbox, Communications Toolbox, Signal Processing Toolbox, Parallel Computing Toolbox)*

### Temporary Demonstrator

University of Sri Jayewardenepura

Colombo, Sri Lanka

Jan 2022 – June 2023

- Worked in the Department of **Electrical and Electronic Engineering** at the University of Sri Jayewardenepura, assisting with *lab preparation*, *practical design*, *teaching support*, *evaluations*, and *workshops*.
- Gained hands-on experience across key areas including *electronics*, *communication systems*, *embedded systems*, *signal processing*, *automation*, *IoT*, and *power systems*.
- Programmed **ATmega328P** and **ESP32** microcontrollers, simulated logic circuits, and configured static/routing protocols, **VLANs**, and **DHCP/DNS** on **Cisco** hardware.
- **Tools & Systems Exposure:** *Arduino UNO, ESP32, Raspberry Pi, Atmel Studio, Proteus, Cisco Packet Tracer, NI Multisim, DIALux, Wireshark, VLAN, TCP/IP*

### Freelance Engineer

Self-Employed (Fiverr)

Remote

Sep 2022 – Aug 2023

- Developed a dynamic web-based **IoT Visualization Interface** for real-time sensor data display on **Google Maps**, integrating **MQTT** and **Node-RED** for live map updates and marker control.
- **Tools Used:** *Node-RED, Google Maps API, JavaScript, MQTT, Docker*
- Designed a coordinated system integrating a differential drive **mobile robot** and a **3-DoF RRR manip-**

ulator for autonomous object transfer, implementing **GPS**-based navigation and **inverse kinematics** for pick-and-place tasks.

- **Tools Used:** *Webots, MATLAB (Robotics System Toolbox)*

## Telecommunication Engineer Intern

*Sri Lanka Telecom PLC*

*Colombo, Sri Lanka*

*Feb 2020 – Aug 2020*

- Worked in the *Planning, Maintenance, and Network (Switching and Transmission)* divisions.
- Gained hands-on experience in *fiber splicing, OTDR-based verification, and fault diagnosis*.
- Assisted with *fiber splicing (fusion/mechanical), ADSL/FTTH router configuration*, as well as the **WDM/MPLS Layer 2/3 VPN** provisioning, maintenance of transmission and switching equipment, including **MSAN** and **eNodeB** infrastructures.
- **Tools & Systems Exposure:** *Fusion Splicer (Sumitomo T-400S), OTDR, Power Meter (FPM-300), MSAN, Clarity OSS, TCP/IP, MPLS, NGN*

## Education

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### University of Kassel

*Master of Science in Electrical Communication Engineering*

*Kassel, Germany*

*Oct 2023 – Present*

- **Current Grade:** 1.68
- **Thesis:** Pulse Design for Pre-6G Mobile Radio Systems
- **Selected Coursework:** *Mobile Radio, Digital Communications, Information Theory & Coding, Signal Detection and Estimation, Simulation of Digital Communication Systems Using MATLAB, MAC Protocols in Wireless Communications,*

### University of Sri Jayewardenepura

*Bachelor of Science of Engineering (Honours) in Electrical and Electronics (minor in Telecommunication Engineering)*

*Colombo, Sri Lanka*

*Nov 2016 – Jun 2021*

- **Grade :** 3.61 (out of 4.0) | equivalent to 1.5 in German Education System
- Second Class Honours (Upper Division)
- **Thesis:** Versatile Indoor Positioning System using Wi-Fi and Deep Learning
- **Selected Coursework:** *Embedded Systems, Digital Electronics, Data Communication, Wireless Communication, Signal Processing, IoT, Communications, Machine Learning, Image Processing, Next Generation Networks, Robotics, Mobile Application Development,*

## Technical Skills

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**Programming Languages:** *C++, Python, MATLAB, Java, JavaScript, Verilog*

**Databases:** *SQLite, MySQL*

**Software / Tools:** *Android Studio, Arduino IDE, Atmel Studio, AutoCAD, Cisco Packet Tracer, DIALux, Docker, L<sup>A</sup>T<sub>E</sub>X, NI Multisim, Node-RED, Octave, OrCAD, Proteus, Vivado, Webots, Wireshark*

## Projects

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### Pulse Design for Pre-6G Mobile Radio Systems

*Oct 2025 – Present*

- Implemented link-level simulations of four candidate waveforms (CP-OFDM, DFT-s-OFDM, FDSS-enhanced DFT-s-OFDM, OTFS) across AWGN and 3GPP fading channels (TDL-D, TDL-C) using **MMSE equalization** under a perfect **channel state information (CSI)** assumption.
- Validated optimal FDSS design over  $5 \times 10^6$  simulation slots per configuration; FDSS achieved **PAPR** 5.81 dB below CP-OFDM baseline while satisfying **ACLR**  $\geq 30$  dBc and **throughput**  $\geq 10$  Mbps.
- **Tools Used:** *MATLAB (5G NR Toolbox, Communications Toolbox, Signal Processing Toolbox, Parallel Computing Toolbox)*

### SAE-CNN Based Indoor Positioning System

*Apr 2025 – Aug 2025*

- A deep learning-based indoor positioning system using Wi-Fi fingerprints, featuring a **stacked autoencoder (SAE)** for feature compression and a **1D-CNN** for classification.

- Applied powered normalization and uniform spatial sampling to improve signal stability and model generalization across indoor environments.
- **Tools Used:** *MATLAB (Deep Learning Toolbox), UJIIndoorLoc, and Tampere datasets*

### Versatile Indoor Positioning System using Wi-Fi and Deep Learning

Oct 2020 – Jun 2021

- A cost-effective, versatile system for accurate smartphone position estimation in an indoor environment using **Wi-Fi RSSIs** and **accelerometer data**.
- The system leverages a **shallow neural network** to classify user location from preprocessed **Wi-Fi** signals, incorporating **accelerometer**-derived motion features to improve input stability and compensate for user movement.
- **Tools Used:** *Python (TensorFlow, Keras), Android Studio*

### Autonomous Search Rescue System

Oct 2020 – Feb 2021

- Proposed a conceptual, cost-effective **autonomous search and rescue** system architecture to improve marine search and rescue operations, particularly for drowning victims, as part of an academic systems design study.
- Outlined a coordinated architecture in which a **UAV** tracks the victim, deploys a flotation device, and guides an **autonomous** rescue vessel, with **GPS** technology facilitating communication between components.

### Gesture-Controlled Smart Room Kit

Mar 2020 – Jun 2021

- A cost-effective solution to convert conventional electric appliances into smart devices, integrating **Wi-Fi** technology within the **smart home** concept.
- This product allows users to control existing home appliances via a smartphone, eliminating the need to replace entire appliances.
- **Tools Used:** *NodeMCU (ESP8266), Android Studio*

### Root-Finding IC Design

Nov 2019 – Jan 2020

- A digital IC designed for root-finding operations using the **IEEE-754 16-bit** floating-point format, featuring a **floating point unit (FPU)** and **master algorithm unit (MAU)**.
- Implemented arithmetic operations and developed hardware logic for **ln(x)** approximation using a 5th-order **Chebyshev polynomial** (Horner method evaluation), including normalization and internal data flow control.
- **Tools Used:** *Verilog HDL*

## Language Skills

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**English:** Fluent (C1)

**German:** Elementary (A2)

**Sinhala:** Native

## Interests

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Embedded Systems | IoT | Robotics and Automation | Wireless Communication | Deep Learning  
| Machine Learning

## Declaration

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I herewith declare that the details provided above are true and accurate to the best of my knowledge.

Ingolstadt, June 30, 2026



Janindu Bandara